Independent monic division review

Seventeen main declarations, nine producer controls and six independent controls passed fresh isolated checking. Thirty-two exact named axiom readbacks use standard Lean axioms. All eight native module exits are zero. A deliberate false equality returns native/gate exit one.

Source correspondence. Quotient assignment overwrites coefficient j; freshness is derived from strictly decreasing remainder length. This justifies the additive quotient update and conservation of $QB + R$. The loop uses the retained word-remainder step and derives success from sufficient fuel.

Exact quotient. The interface starts from an independently supplied factorization, derives existence of an empty-remainder return and then uniqueness by cancellation. It does not assume the desired return value. Prime modulus, canonical words, $p < 2^{31}$ and literal monic last coefficient are explicit premises.

Independent controls. Zero-word divisor rejection, degree-short input with zero fuel, constant-one divisor, wrong quotient sign/degree and nonfresh overwrite passed. Producer controls also test insufficient fuel and exact/nonexact two-step outputs.

Boundary. The quotient is a polynomial accumulator with assignment laws, not a full native Vec simulation. Allocation/storage/trim correspondence, usize allocation overflow, nonmonic inversion and general pegcd remain separate. No scientific measurements or source mutations occurred.

Custody. Published sources and prospective plan were bound; source-first obligations preceded proof inspection. Historical monic/helper familiarity is disclosed. An initial dependency-copy layout failure was retained and corrected. Cached dependencies and compiler remain trusted; no independent binary kernel check ran.